

Experiment 22: Perform Sharpening of Image using Laplacian Mask with Positive Center Coefficient

AIM

To sharpen an image using a Laplacian mask with a positive center coefficient.

PROGRAM

```
import cv2

import numpy as np

# Read the image

img = cv2.imread(r"C:\Users\chait\OneDrive\Documents\Desktop\cv lab image.jpg")

# Laplacian mask with positive center coefficient

kernel = np.array([[0, -1, 0],
                   [-1, 5, -1],
                   [0, -1, 0]])

# Apply filter

sharp = cv2.filter2D(img, -1, kernel)

# Display images

cv2.imshow("Original Image", img)

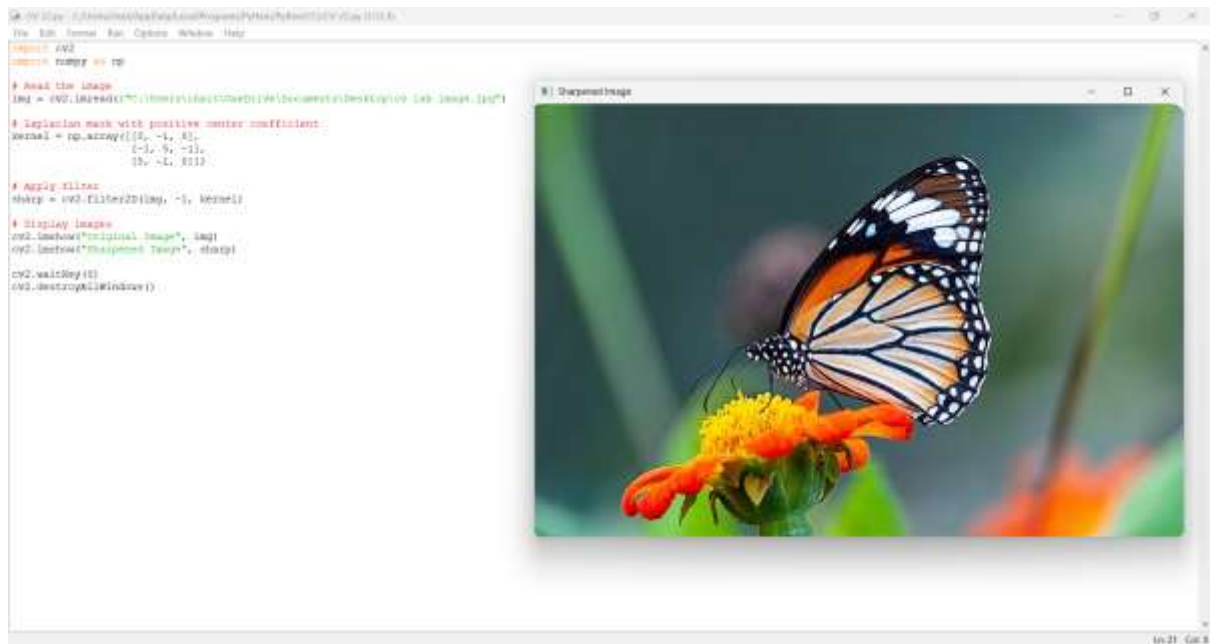
cv2.imshow("Sharpened Image", sharp)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

OUTPUT

The sharpened image is displayed after applying the positive center Laplacian mask.



Explanation

We applied a sharpening filter with a positive center coefficient using a Laplacian mask. The filter enhances object boundaries and improves the overall clarity of the image.